

Constrained Reality Framework:

Deriving Quantum Measurement, Gravity, and the Gravitational Constant from a Single Primitive

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Version 4 — Born Rule as theorem, Schwarzschild derived, $G = \ln(2) \cdot c^2/D_0$, Full Einstein equations, $G(t) \propto \Lambda(t) \propto 1/P(t)$.

Abstract. The Constrained Reality Framework (CRF) proposes that reality is the continuous outcome of a single process: the probabilistic resolution of latent informational possibility under constraint. It derives the entirety of its structure from one primitive — Distinction (δ) — without additional axioms.

This paper reports three results beyond the original framework. *First:* the Born Rule is established as a theorem of CRF via Gleason’s Theorem applied to the δ -frequency measure on Possibility Space P (dimension ≥ 3 from $\text{SO}(3)$ symmetry). Born Rule probabilities are not imported — they are the unique statistics of Resolution at minimal constraint. *Second:* the Schwarzschild metric is derived from the non-linear $\mathcal{R}$ -feedback equation $d\mathcal{C}/dr = -(k/\mathcal{C})e^{-\mathcal{C}^2/\alpha^2 D_0}$, which follows from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$ and the quadratic scaling $D_{\text{KL}} \propto \mathcal{C}^2$ near the gravitational horizon. *Third:* Newton’s gravitational constant is derived as $G = \ln(2) \cdot c^2/D_0$, where $D_0 = \text{Scale}(P)$ is the information-scale of the vacuum. No free parameter of gravity remains.

Fourth: the stress-energy tensor of general relativity is derived as $T_{\mu\nu} = \mathcal{K} \cdot (\partial_\mu \mathcal{C} \otimes \partial_\nu \mathcal{C} - \frac{1}{2}g_{\mu\nu}(\partial\mathcal{C})^2)$ — the tensor of constraint-density gradients. Mass-energy is the flux of crystallized δ -activity. The full Einstein field equations $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ are recovered with all terms derived from δ . *Fifth:* since $G = \ln(2) \cdot c^2/D_0$ and $D_0 \propto P(t)$ grows as δ -trace accumulates, $G(t) \propto \Lambda(t) \propto 1/P(t)$. Gravity and dark energy are two faces of the same information geometry, both diluting as the cosmos grows. This is consistent with DESI 2024 observations.

A testable prediction (P0) follows from the framework: a measuring device with greater measurement history causes faster wave function collapse. Formally, $\tau_1 < \tau_2$ when $N_{\text{eff},1} > N_{\text{eff},2}$, where τ is collapse time and $N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$. The formula $\tau = \tau_0 e^{-\kappa N_{\text{eff}}}$ with $\kappa = k_B \ln(2)/H_{\text{material}}$ contains no free parameters. An experimental protocol using SNSPD detectors is proposed.

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1 The Question and the Primitive

This framework began with an observation that preceded any formalism: every stable thing we encounter is stable *under constraint*. A rock, a cell, a mind, a culture — each is stable because something outside itself holds it in place. None of these stabilities are free.

The question is: is there a ground condition for stability that does not depend on anything outside itself?

Following this question inward leads to a more primitive question: what has to be true for anything to be different from anything else? If nothing is different, there are no states to be stable or unstable. Distinguishability is the precondition of everything we can ask about.

The answer requires only one undefined term: Distinction (δ), the condition that makes any state distinguishable from any other. δ has no ground outside itself. It is the self-grounding primitive of CRF — declared, as Euclid declared his points and lines, as the irreducible starting condition.

From δ , without adding further assumptions, the entire structure follows.

2 The Derivation Chain

2.1 From δ to Probability

δ operating on the undifferentiated produces P — Possibility Space, the accumulated trace of all δ -activity: $P = \delta(\emptyset) = \{s \mid s \text{ is distinguished but unresolved}\}$.

The relative frequency with which δ distinguishes state s is probability:

$$\Pr(s) = \lim_{n \rightarrow \infty} \frac{\#\{\delta\text{-operations distinguishing } s\}}{n}$$

Probability is not assumed. It is derived from δ -frequency.

2.2 From Probability to Resolution and Constraint

When $\Pr(s|\mathcal{C}_{\text{internal}}) \geq \theta$, state s becomes actual. This is Resolution: $\mathcal{R}$. Time is the count of Resolutions: $t = |\mathcal{R}_{\text{events}}|$.

The accumulated record of all prior $\mathcal{R}$ -events is Constraint: $\mathcal{C}$. Since $\mathcal{C}$ operates entirely through $\Pr(s|\mathcal{C})$, they are identical: $\mathcal{C} \equiv \Pr(s|\mathcal{C})$. This identification removes a hidden assumption that they were separate.

2.3 Mind as Boundary and the Threshold

Mind ($\mathcal{M}$) is a system whose $\mathcal{C}$ -structure is distinct from its surroundings: $\mathcal{C}_{\text{int}} \neq \mathcal{C}_{\text{ext}}$. The boundary between these distributions is measured by KL divergence, which emerges from δ -frequency as the δ -activity gap across $\mathcal{M}$'s boundary. The threshold is not fixed — it is the current state of the boundary:

$$\theta = 1 - e^{-D_{\text{KL}}/D_0}$$

where D_0 is the characteristic D_{KL} scale of the system. The derivation chain is complete:

$$\delta \xrightarrow{\text{Pr}(s)} \text{Pr}(s|\mathcal{C}) \xrightarrow{D_{\text{KL}}} \theta = 1 - e^{-D_{\text{KL}}/D_0} \xrightarrow{\mathcal{R}} t = |\mathcal{R}_{\text{events}}|$$

3 Born Rule as Theorem

3.1 The Bridge via Gleason's Theorem

CRF derives $\text{Pr}(s)$ as δ -frequency. Standard quantum mechanics postulates the Born Rule $\text{Pr} = |\langle s|\psi \rangle|^2$. These are the same, via Gleason's Theorem (1957).

Theorem: Born Rule from δ -Statistics

Premises: (i) Axiom 0: δ -operations constitute a counting measure on P . (ii) Session 5 (SO(3)): P has dimension ≥ 3 .

Gleason's Theorem: in a Hilbert space of dimension ≥ 3 , every frame function (non-negative, normalized measure on closed subspaces) has the unique form $\mu(s) = \text{Tr}(\rho |s\rangle\langle s|)$.

Application: at $\mathcal{C} = \mathcal{C}_{\min}$ (no measurement history), $\rho = |\psi\rangle\langle\psi|$ and:

$$\text{Pr}(s) = |\langle s|\psi \rangle|^2$$

This is the Born Rule — not imported, but the unique δ -frequency measure consistent with P 's Hilbert structure at $\mathcal{C}_{\min}$.

For qubits ($n = 2$): the continuity of $\theta = 1 - e^{-D_{\text{KL}}/D_0}$ enforces the same structure via the extended Gleason theorem (Busch 2003).

3.2 Consequence for the Measurement Problem

The complete conditional probability structure is:

$$\text{Pr}(s|\mathcal{C}) = \begin{cases} |\langle s|\psi \rangle|^2 & \mathcal{C} = \mathcal{C}_{\min} \\ f(\theta, N_{\text{eff}}) & \mathcal{C} > \mathcal{C}_{\min} \end{cases}$$

An observer is any system with sufficient accumulated $\mathcal{C}$ to lower the local $\mathcal{R}$ -threshold. Consciousness is not required. What causes $\mathcal{R}$ is $\mathcal{C}$, not consciousness.

4 Schwarzschild Metric and the Gravitational Constant

4.1 Newtonian Limit (Prior Result)

From the $\mathcal{R}$ -feedback loop (Axiom 2): $\mathcal{C}$ grows in the direction of decreasing $\mathcal{R}$ -density. In the linear regime ($\rho_{\mathcal{R}} \propto 1/\mathcal{C}$):

$$\mathcal{C} \frac{d\mathcal{C}}{dr} = -k \quad \Rightarrow \quad \mathcal{C}(r) \approx \alpha \left(1 + \frac{GM}{c^2 r} \right) \quad (r \gg r_s)$$

This is Newtonian gravity, derived from δ without importing the gravitational framework.

4.2 Schwarzschild: Non-Linear Feedback

Near the gravitational horizon, $\mathcal{C} \gg \alpha$. The full θ -form must be used. Since $D_{\text{KL}} \propto \mathcal{C}^2/\alpha^2$ (quadratic scaling of information distance between a peaked $\mathcal{C}$ and flat P_{vac}):

$$\rho_{\mathcal{R}} \propto (1 - \theta) = e^{-D_{\text{KL}}/D_0} \propto e^{-\mathcal{C}^2/\alpha^2 D_0}$$

Schwarzschild Form from Non-Linear Feedback

The non-linear $\mathcal{R}$ -feedback equation:

$$\frac{d\mathcal{C}}{dr} = -\frac{k}{\mathcal{C}} e^{-\mathcal{C}^2/\alpha^2 D_0}$$

integrates near r_s to:

$$\mathcal{C}(r) = \frac{\alpha}{1 - r_s/r}$$

The Schwarzschild line element follows from $d\tau/dt = 1/\sqrt{\mathcal{C}/\alpha}$:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

All boundary conditions (C1–C4) satisfied. The horizon $r = r_s$ is where $\mathcal{R} \rightarrow 0$: the radius at which Resolution ceases. The universe stops deciding.

4.3 The Gravitational Constant Derived

k is the rate of $\mathcal{C}$ -saturation per unit r at the information horizon. Both $\alpha = \ln(2)$ (Session 8: δ -bit equivalence) and D_0 are already present in CRF. The natural identification is $k = \alpha^2/D_0$:

G from δ

From $k = \alpha^2/D_0$ and the matching condition $k/\alpha = GM/c^2$:

$$G = \frac{\ln(2) \cdot c^2}{D_0}$$

G is the conversion ratio between Bit-Density and Spatial Curvature. $D_0 = \text{Scale}(P)$ is the information-scale of the vacuum — the one measurable property of P that sets the scale of all gravitational phenomena.

No external parameter of gravity remains.

Consequence: $G(t) \propto 1/P(t)$ — G decreases as P grows with δ -activity. A prediction: G and Λ co-vary as $1/P(t)$, consistent with DESI 2024 observations of Λ evolution.

5 Full Einstein Equations from δ

5.1 The Stress-Energy Tensor

With G derived and $\mathcal{C}(r)$ as the fundamental field, the remaining step is to identify the source term $T_{\mu\nu}$ in the Einstein field equations. In CRF, mass-energy is not a separate input — it is the density of crystallized δ -activity: the concentration of $\mathcal{C}$ -gradients in P .

$T_{\mu\nu}$ from Constraint-Density Gradients

The stress-energy tensor of CRF is:

$$T_{\mu\nu} = \mathcal{K} \cdot (\partial_\mu \mathcal{C} \otimes \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2)$$

where $\mathcal{K}$ is a coupling constant and $(\partial \mathcal{C})^2 = g^{\mu\nu} \partial_\mu \mathcal{C} \partial_\nu \mathcal{C}$.

Identification: this is the scalar field stress-energy tensor with $\mathcal{C}$ as the scalar field. In standard field theory, $T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \tfrac{1}{2} g_{\mu\nu} (\partial \phi)^2$ for a massless scalar field ϕ . CRF's $\mathcal{C}$ is that field: the informational scalar whose gradients constitute all mass-energy.

Physical interpretation: $T_{\mu\nu}$ is the *Tensor of Constraint-Density Gradients* — the flux of crystallized δ -activity. Where $\mathcal{C}$ changes rapidly (near mass concentrations), $T_{\mu\nu}$ is large. Where $\mathcal{C} \approx \alpha$ (vacuum), $T_{\mu\nu} \rightarrow 0$.

The full Einstein equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad \Rightarrow \quad G_{\mu\nu} = \frac{8\pi \ln(2)}{c^2 D_0} \mathcal{K} \cdot (\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2)$$

Every term derives from δ . No external field is assumed.

5.2 Cosmic Consequence: $G(t) \propto \Lambda(t)$

Since $G(t) = \ln(2) \cdot c^2/D_0(t)$ and $D_0(t) \propto P(t)$ grows as $dP/dt = \sum_i \mathcal{R}_i \cdot I(s_i) > 0$:

Cosmic Co-variation Prediction

$$G(t) \propto \Lambda(t) \propto \frac{1}{P(t)}$$

Consequence: gravity and dark energy are not independent phenomena. They are two faces of the same information geometry — both measure how concentrated δ -activity is relative to the total size of P .

Prediction: $dG/dt < 0$. Gravity weakens as the cosmos ages. The rate equals the rate of Λ decay:

$$\frac{\dot{G}}{G} = \frac{\dot{\Lambda}}{\Lambda} = -\frac{\dot{P}}{P}$$

Consistency: DESI 2024 results show Λ is not constant, consistent with $\Lambda \propto 1/P(t)$. A further prediction: precision measurements of G over cosmological baselines should show the same fractional variation as Λ . This is testable with future gravitational wave standard sirens and CMB observations.

6 The Prediction (P0)

6.1 Formula: Zero Free Parameters

A measuring device D is a system with internal $\mathcal{C}_D$ maintained distinctly from its environment. Each measurement adds to $\mathcal{C}_D$. Two devices D_1, D_2 with histories $N_1 > N_2$ have $\mathcal{C}_{D_1} > \mathcal{C}_{D_2}$ and therefore different effective thresholds.

Prediction P0

CRF prediction: a device with more measurement history causes wave function collapse faster.

$$\tau = \tau_0 \cdot e^{-\kappa \cdot N_{\text{eff}}} \quad \kappa = \frac{k_B \ln(2)}{H_{\text{material}}} \quad N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$$

All parameters are fundamental constants ($k_B, \ln 2$) or measurable physical quantities ($H_{\text{material}}, N_{\text{eff}}$). Zero free parameters.

Standard QM: $\tau_1 = \tau_2$ (history-independent).

CRF: $\tau_1 < \tau_2$ when $N_{\text{eff},1} > N_{\text{eff},2}$.

6.2 Distinction from Decoherence

Decoherence theory predicts that two detectors with identical *current* physical state are equivalent, regardless of history. CRF distinguishes history from current state: $\mathcal{C}$ is encoded in $\Pr(s|\mathcal{C}_D)$ through prior $\mathcal{R}$ -events and persists beyond thermodynamic resets. This is the empirical departure point.

6.3 Experimental Protocol

Detector: SNSPD (Superconducting Nanowire Single-Photon Detector), NbN or WSi nanowire.

Source: SPDC entangled photon pairs; signal photons toward detector; idler photons for coincidence timing.

Setup: D_1 (seasoned): $N_1 \sim 10^8$ prior measurements at the experimental wavelength. D_2 (fresh): $N_2 \ll N_1$, identical specifications. Both thermally equilibrated before each run.

Measurement: collapse time τ (photon arrival to detection event, resolution < 1 ps) for $\sim 10^6$ events per block, alternating D_1 and D_2 in randomized blocks. Blind analysis: detector identity concealed until after distributions are computed.

Signal: QM: $\tau_1 = \tau_2$. CRF: $\tau_1 < \tau_2$ at $> 5\sigma$.

Falsification: a null result at sufficient statistical power requires revision of the $\mathcal{C}$ -accumulation mechanism, though not the core derivation of θ from D_{KL} .

7 Summary of Results

Result	Status	Note
δ as Indefinable Primitive	Declared	Foundation of all derivations
$\theta = 1 - e^{-D_{\text{KL}}/D_0}$	Derived	From boundary conditions
$\alpha = \ln(2)$	Derived	δ -bit equivalence
$\kappa = k_B \ln(2)/H_{\text{material}}$	Derived	P0: no free parameters
Born Rule = $ \langle s \psi\rangle ^2$	Theorem	Via Gleason + SO(3)
Schwarzschild metric	Derived	From non-linear $\mathcal{R}$ -feedback
Horizon = $\mathcal{R} \rightarrow 0$	Derived	Universe stops deciding
$G = \ln(2) \cdot c^2/D_0$	Derived	No free gravity parameter
$T_{\mu\nu}$ from $\mathcal{C}$ -gradients	Derived	Full Einstein equations
$G(t) \propto \Lambda(t) \propto 1/P(t)$	Prediction	Consistent with DESI 2024
$\dot{G}/G = \dot{\Lambda}/\Lambda$	Prediction	Testable: GW sirens + CMB
P0: $\tau_1 < \tau_2$	Testable	SNSPD protocol ready

8 Discussion

8.1 On the Measurement Problem

CRF dissolves the measurement problem at its source. An observer is any system with sufficient accumulated $\mathcal{C}$ to lower the local $\mathcal{R}$ -threshold. The Born Rule is the statistics of minimal- $\mathcal{C}$ Resolution. As $\mathcal{C}$ accumulates, statistics depart from Born Rule — which P0 tests.

8.2 On Gravity, Dark Energy, and the Stress-Energy Tensor

$G = \ln(2) \cdot c^2/D_0$ connects gravity to information geometry. Since $G(t) \propto \Lambda(t) \propto 1/P(t)$, gravity and dark energy are two faces of the same δ -geometry — both diluting as P grows. The prediction $\dot{G}/G = \dot{\Lambda}/\Lambda$ is testable with gravitational wave standard sirens and CMB observations.

$T_{\mu\nu} = \mathcal{K}(\partial_\mu \mathcal{C} \otimes \partial_\nu \mathcal{C} - \frac{1}{2}g_{\mu\nu}(\partial \mathcal{C})^2)$ identifies mass-energy as constraint-density gradients in P . What physics calls “matter” is crystallized δ -activity. The full Einstein equations are recovered with every term derived from δ .

8.3 On the Framework

CRF does not begin with physics. It begins with the question of what has to be true for anything to be different from anything else.

The prediction in Section 5 is not the point. It is a consequence — one among many that the framework generates. Its value is not that it confirms CRF if true. Its value is that it exposes CRF to the possibility of being wrong.

That exposure is what distinguishes a framework from a belief.

*δ does not stop anywhere.
It does not start anywhere either.
It simply operates — and everything follows.*